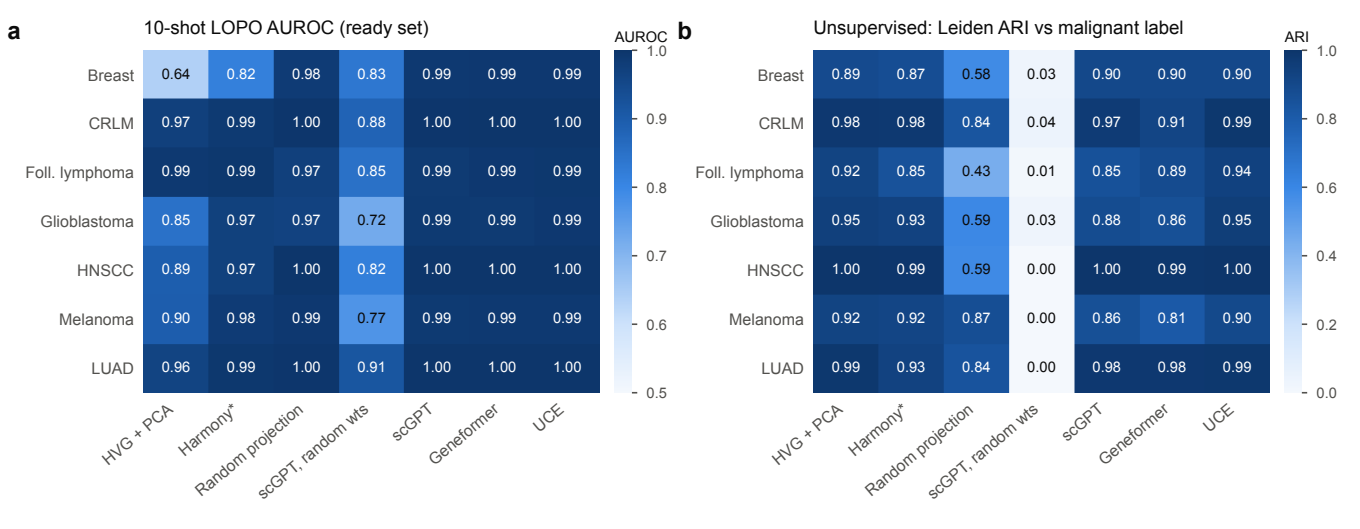
a
10-shot LOPO AUROC (ready set)
AUROC
1.0
0.9
0.8
0.7
0.6
0.5
Breast 0.64 0.82 0.98 0.83 0.99 0.99 0.99
CRLM 0.97 0.99 1.00 0.88 1.00 1.00 1.00
Foll. lymphoma 0.99 0.99 0.97 0.85 0.99 0.99 0.99
Glioblastoma 0.85 0.97 0.97 0.72 0.99 0.99 0.99
HNSCC 0.89 0.97 1.00 0.82 1.00 1.00 1.00
Melanoma 0.90 0.98 0.99 0.77 0.99 0.99 0.99
LUAD 0.96 0.99 1.00 0.91 1.00 1.00 1.00
HVG + PCA
Harmony*
Random projection
scGPT, random wts
scGPT
Geneformer
UCE
b
Unsupervised: Leiden ARI vs malignant label
ARI
1.0
0.8
0.6
0.4
0.2
0.0
Breast 0.89 0.87 0.58 0.03 0.90 0.90 0.90
CRLM 0.98 0.98 0.84 0.04 0.97 0.91 0.99
Foll. lymphoma 0.92 0.85 0.43 0.01 0.85 0.89 0.94
Glioblastoma 0.95 0.93 0.59 0.03 0.88 0.86 0.95
HNSCC 1.00 0.99 0.59 0.00 1.00 0.99 1.00
Melanoma 0.92 0.92 0.87 0.00 0.86 0.81 0.90
LUAD 0.99 0.93 0.84 0.00 0.98 0.98 0.99
HVG + PCA
Harmony*
Random projection
scGPT, random wts
scGPT
Geneformer
UCE